

Leave Management System API Documentation

Version

1.0

Base URL

```
{{baseUrl}}
```

Authentication

Most endpoints require JWT Bearer Authentication.

Header

```
Authorization: Bearer <accessToken>
```

1. Authentication APIs

1.1 Register User

Endpoint

```
POST /api/users/auth/register
```

Description

Registers a new employee account in the system.

Request Body

```
{
  "email": "user@example.com",
  "password": "12345678",
  "username": "user123",
  "name": "John Doe",
```

```
{
  "managerId": "11111111-1111-1111-1111-111111111111"
}
```

Success Response

201 Created

```
{
  "message": "User registered successfully"
}
```

Error Responses

User Already Exists

409 Conflict

```
{
  "error": "User already exists"
}
```

1.2 Login

Endpoint

```
POST /api/users/auth/login
```

Description

Authenticates a user and returns access credentials.

Request Body

```
{
  "email": "manager@company.com",
  "password": "manager123"
}
```

Success Response

200 OK

```
{
  "accessToken": "<jwt-token>",
  "user": {
    "id": "uuid",
    "email": "manager@company.com",
    "role": "MANAGER"
  }
}
```

Error Responses

User Not Found

401 Unauthorized

```
{
  "error": "No user found with that email"
}
```

Invalid Password

401 Unauthorized

```
{
  "error": "Invalid credentials"
}
```

Rate Limit Exceeded

429 Too Many Requests

```
{
  "error": "Too many login attempts. Try again later."
}
```

1.3 Refresh Token

Endpoint

```
GET /api/users/auth/refresh-token
```

Description

Generates a new access token using a valid refresh token.

Success Response

200 OK

```
{
  "accessToken": "<new-jwt-token>"
}
```

Error Responses

Too Many Requests

429 Too Many Requests

```
{
  "error": "Too many login attempts. Try again later."
}
```

2. User APIs

2.1 Get User Profile

Endpoint

```
GET /api/users/profile
```

Authentication

Required

Success Response

```
{
  "id": "uuid",
  "email": "employee@company.com",
  "name": "John Doe",
  "role": "EMPLOYEE"
}
```

2.2 Get Employees

Endpoint

```
GET /api/users/employees
```

Authentication

Required

Description

Returns a list of employees available to the authenticated manager.

2.3 Update Reporting Manager

Endpoint

```
PATCH /api/users/update/manager
```

Authentication

Required

Request Body

```
{
  "employeeId": "uuid",
  "managerId": "uuid"
}
```

Success Response

```
{
  "message": "Manager updated successfully"
}
```

2.4 Delete User

Endpoint

```
DELETE /api/users/{userId}
```

Authentication

Required

Authorization

Admin Only

Success Response

```
{
  "message": "User deleted successfully"
}
```

3. Leave APIs

3.1 Apply Leave

Endpoint

```
POST /api/leaves/apply
```

Authentication

Required

Request Body

```
{
  "type": "CASUAL",
  "startDate": "2026-06-16",
  "endDate": "2026-06-16",
  "reason": "Family function"
}
```

Success Response

```
{
  "_id": "leaveId",
  "status": "PENDING"
}
```

Business Rules

- Employee must have sufficient leave balance.
 - Leave dates must be valid.
 - Leave request enters PENDING state.
-

3.2 Get Leave Balance

Endpoint

```
GET /api/leaves/balance
```

Authentication

Required

Description

Returns leave balances for the current year.

Success Response

```
{
  "year": 2026,
  "balances": [
    {
```

```
    "leaveType": "CASUAL",
    "allocated": 12,
    "carriedForward": 0,
    "used": 2,
    "remaining": 10
  }
]
```

3.3 Get Leave History

Endpoint

```
GET /api/leaves/history
```

Authentication

Required

Query Parameters

Parameter	Description
status	PENDING, APPROVED, REJECTED, CANCELLED
type	CASUAL, SICK, PRIVILEGE
startDate	Filter start date
endDate	Filter end date
managerId	Manager filter

Example

```
GET /api/leaves/history?status=PENDING&type=CASUAL
```

Success Response

```
{
  "leaves": [
    {
      "_id": "leaveId",
      "employeeId": "uuid",

```



```
    "type": "CASUAL",
    "status": "PENDING",
    "reason": "Family function"
  }
],
"total": 1,
"page": 1,
"limit": 10
}
```

3.4 Get Pending Leave Requests

Endpoint

```
GET /api/leaves/requests/pending
```

Authentication

Required

Authorization

Manager Only

Description

Returns all leave requests awaiting manager approval.

3.5 Approve Leave

Endpoint

```
PATCH /api/leaves/{leaveId}/approve
```

Authentication

Required

Authorization

Reporting Manager Only

Request Body

```
{
  "comment": "Approved"
}
```

Success Response

```
{
  "status": "APPROVED"
}
```

Error Responses

Not Reporting Manager

```
{
  "error": "Unauthorized: You are not the reporting manager"
}
```

3.6 Reject Leave

Endpoint

```
PATCH /api/leaves/{leaveId}/reject
```

Authentication

Required

Authorization

Reporting Manager Only

Request Body

```
{
  "comment": "Project deadline approaching"
}
```

Success Response

```
{
  "status": "REJECTED"
}
```

Error Responses

```
{
  "error": "Unauthorized: You are not the reporting manager"
}
```

3.7 Cancel Leave

Endpoint

```
PATCH /api/leaves/{leaveId}/cancel
```

Authentication

Required

Description

Cancels an existing leave request.

Request Body

```
{
  "comment": "Workload is high"
}
```

Success Response

```
{
  "_id": "leaveId",
  "status": "CANCELLED"
}
```

Error Responses

Cannot Cancel Another User's Leave

```
{
  "error": "Unauthorized: You can only cancel your own leave requests"
}
```

Leave Is Not Pending

```
{
  "error": "Only pending leave requests can be cancelled"
}
```

Error Codes

Status Code	Meaning
200	Success
201	Resource Created
400	Validation Error
401	Unauthorized
403	Forbidden
404	Not Found
409	Conflict
429	Too Many Requests
500	Internal Server Error

Rate Limiting

Authentication endpoints are rate limited.

Example headers:

```
X-RateLimit-Limit: 5
X-RateLimit-Remaining: 0
Retry-After: 146
```

When the limit is exceeded:

```
{
  "error": "Too many login attempts. Try again later."
}
```

Leave Lifecycle

```
PENDING
|
├── APPROVED
├── REJECTED
└── CANCELLED
```

Roles & Permissions

Endpoint	Employee	Manager	Admin
Apply Leave	✓	✓	✓
Leave Balance	✓	✓	✓
Leave History	✓	✓	✓
Approve Leave	✗	✓	✓
Reject Leave	✗	✓	✓
Cancel Own Leave	✓	✓	✓
Delete User	✗	✗	✓
Update Manager	✗	✓	✓